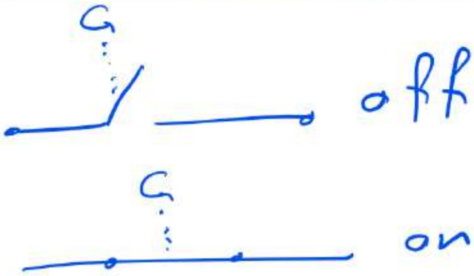
L #1

1819w AISG

Transistor → 1- switch
→ 2 - Resistor



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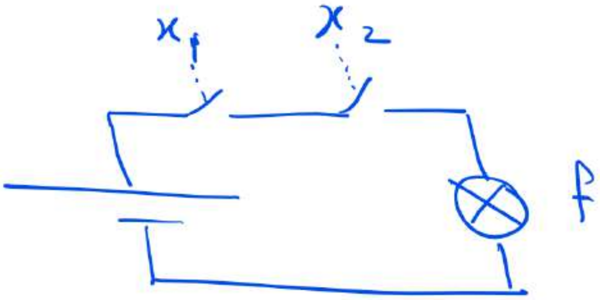
* AND logic
(series)



light

$$f(x_1, x_2) = \underbrace{x_1 . x_2}_{\text{AND}}$$

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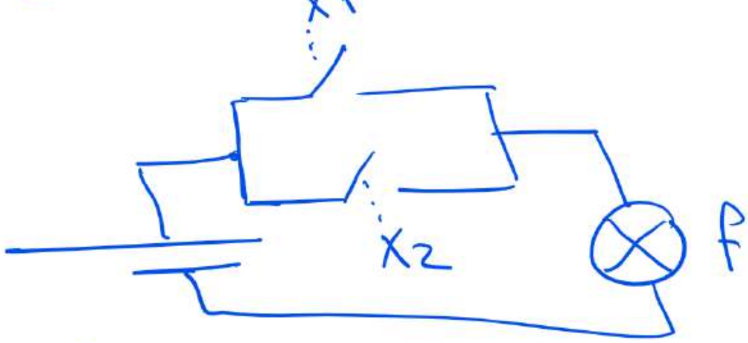
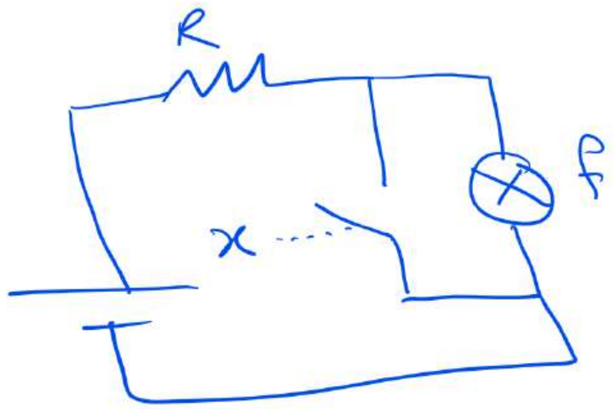
* OR logic (parallel)



$$f(x_1, x_2) = x_1 \underbrace{+}_{\text{OR}} x_2$$

NOT Logic (inverter)



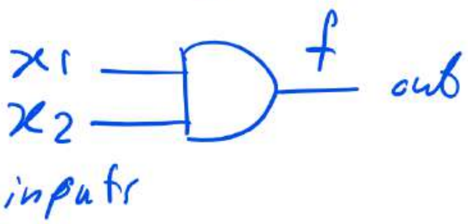
if $x = \underline{1}$ (on) $\longrightarrow f = \underline{0}$ off

$x = \underline{0}$ (off) $\longrightarrow f = \underline{1}$ ON

$$\left. \begin{aligned} f(x) &= \bar{x} \\ &= \sim x \\ &= x' \end{aligned} \right\} \text{NOT}$$

Gate / Truth table

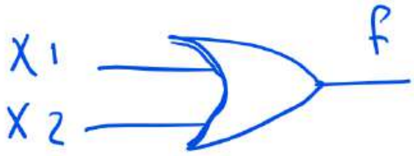
① AND gate



x_1	x_2	f
0	0	0
0	1	0
1	0	0
1	1	1

$$\left. \begin{aligned} f(x_1, x_2) &= x_1 . x_2 \\ &= x_1 x_2 \end{aligned} \right\}$$

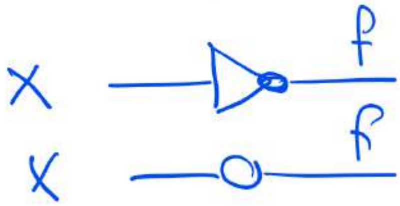
2 OR gate



$$f(x_1, x_2) = x_1 + x_2$$

x_1	x_2	f
0	0	0
0	1	1
1	0	1
1	1	1

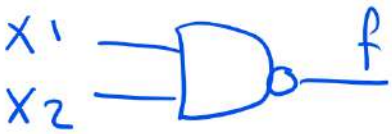
3 NOT gate



x	f
0	1
1	0

$$f(x) = \overline{x}$$

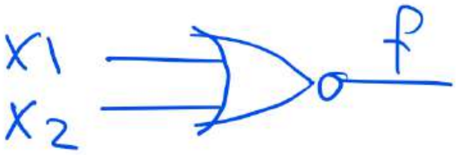
4 NAND (AND-NOT)



$$f(x_1, x_2) = \overline{x_1 \cdot x_2}$$

x_1	x_2	f
0	0	1
0	1	1
1	0	1
1	1	0

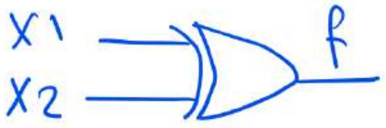
5 NOR gate (OR-NOT)



$$f(x_1, x_2) = \overline{x_1 + x_2}$$

x_1	x_2	f
0	0	1
0	1	0
1	0	0
1	1	0

6) XOR : exclusive OR gate



x_1	x_2	f
0	0	0
0	1	1
1	0	1
1	1	0

مختلفين

$$f(x_1, x_2) = x_1 \oplus x_2$$
$$= \underline{\overline{x_1}} \cdot \underline{x_2} + \underline{x_1} \cdot \underline{\overline{x_2}}$$

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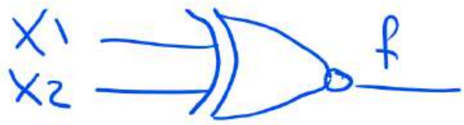
7) XNOR : XOR - NOT



x_1	x_2	f
0	0	1
0	1	0
1	0	0
1	1	1

$$f(x_1, x_2) = \overline{x_1 \oplus x_2}$$
$$= \underline{x_1} \cdot \underline{x_2} + \underline{\overline{x_1}} \cdot \underline{\overline{x_2}}$$

XNOR

$$x_1 \odot x_2 = \overline{x_1 \oplus x_2}$$
$$= \underline{\overline{x_1}}\, \underline{\overline{x_2}} + \underline{x_1}\, \underline{x_2}$$

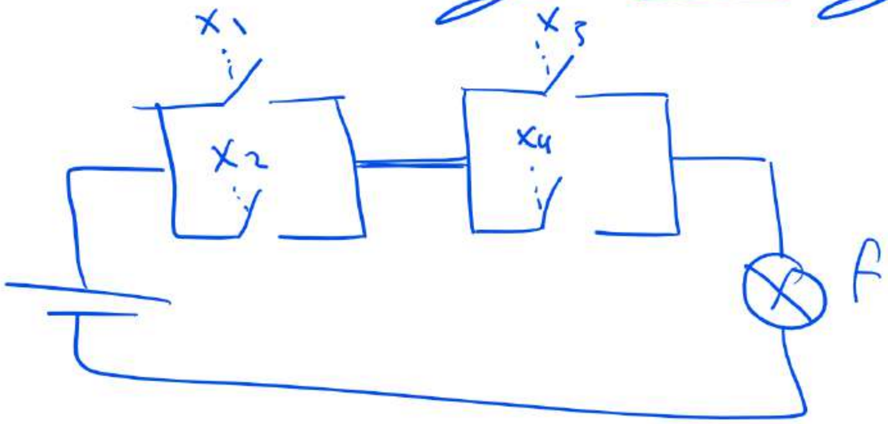
# of rows $= 2^n$; n = # of variables (inputs)

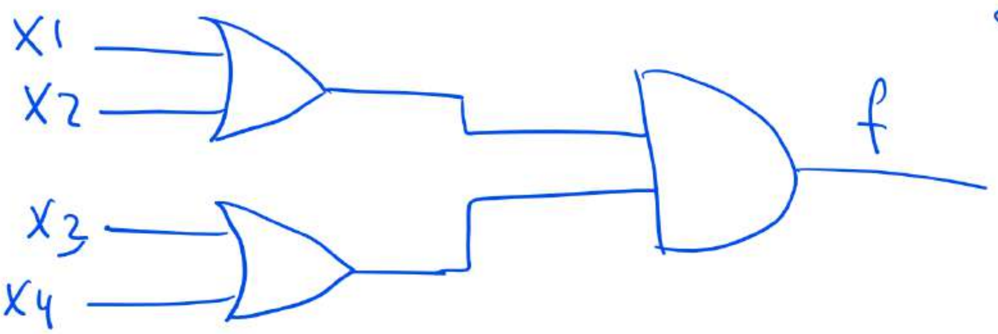
$n = 2 \rightarrow row = 2^2 = 4$

$n = 3 \rightarrow row = 2^3 = 8$

Truth table: is used to show the functional behavior of the circuit.

Draw the following circuit using gates

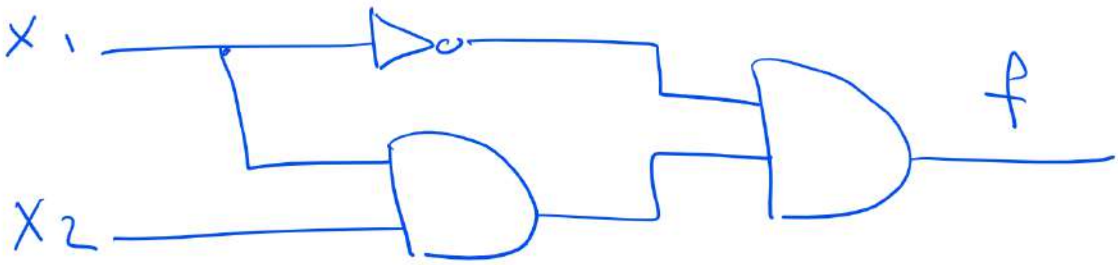




Network of gates

$$f(x1, x2, x3, x4) = (x1+x2)\cdot(x3+x4)$$

Example I: Given the following

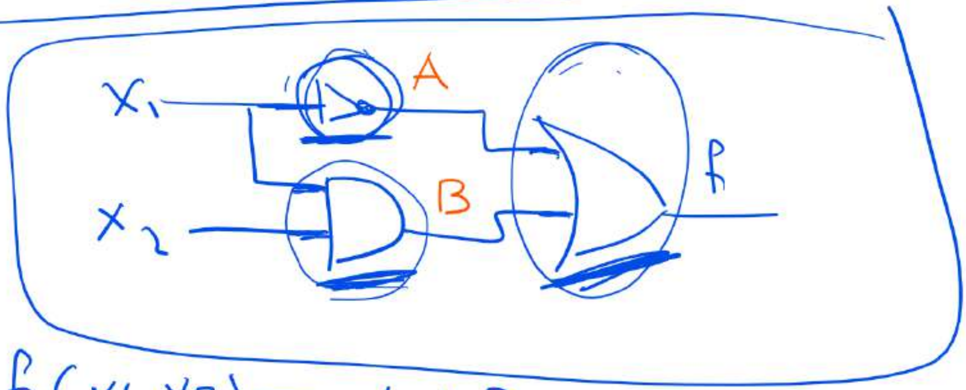


1_ what is the function of the circuit
→ Logical expression

2_ Draw truth table?

3- what is the cost of the circuit?

4- Draw timing diagram?



$$f(x_1, x_2) = A + B$$

$$= \overline{x_1} + B$$

$$\boxed{f = \overline{x_1} + x_1 . x_2}$$

2- Truth table

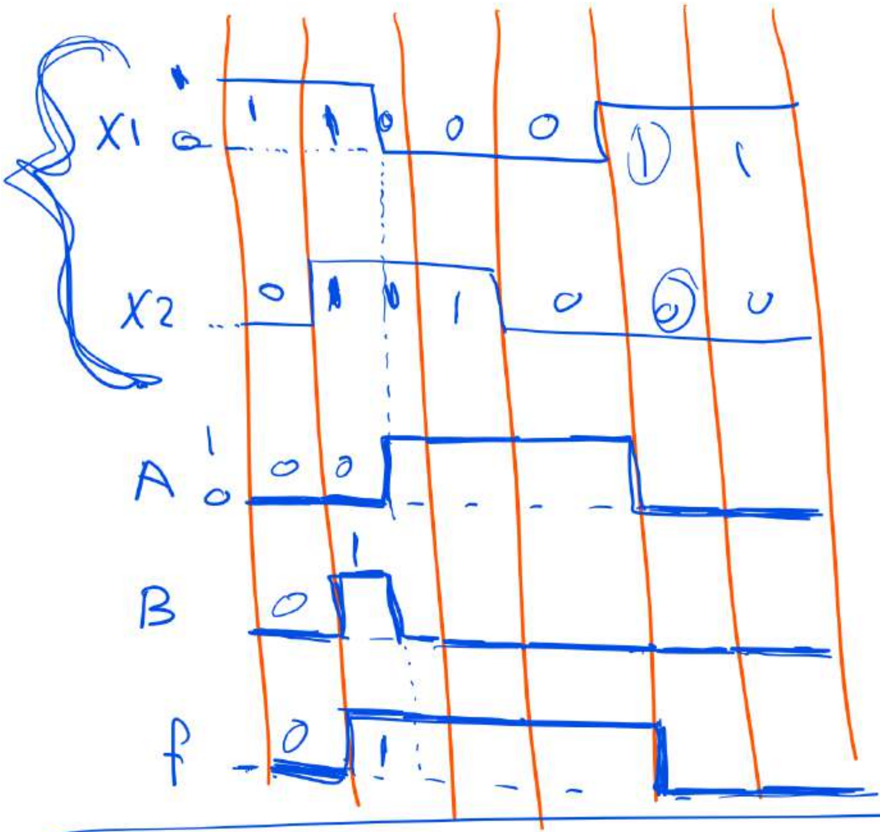
x_1	x_2	A	B	f
0	0	1	0	1
0	1	1	0	1
1	0	0	0	0
1	1	0	1	1

3- Cost: # of gates
+ inputs for gates

x_1	x_2	f
0	0	0
0	1	1
1	0	1
1	1	1

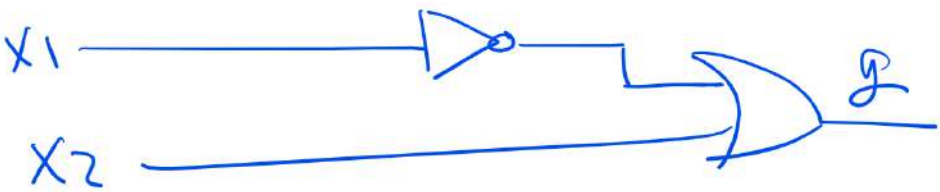
$$\text{cost} = 3 + \underline{5} = 8$$

4- Timing diagram:



x_1	x_2	A	B	f
0	0	1	0	1
0	1	1	0	1
1	0	0	0	0
1	1	0	1	1

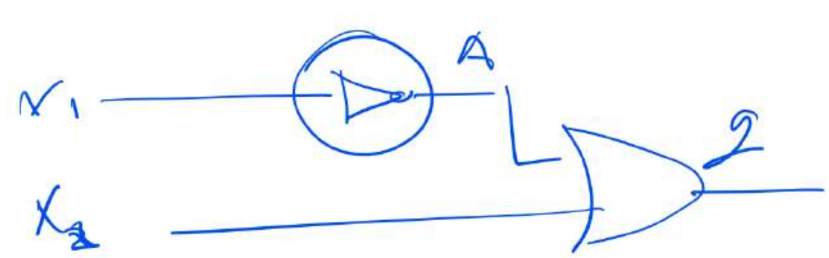
Example II: Given the following circuit



1- Derive the logical expression

2- Draw truth table

3- What is the cost of the circuit?



1- $g(x_1, x_2) = A + x_2$

$$\boxed{g = \overline{x_1} + x_2}$$

2- Truth table

x_1	x_2	A	g
0	0	1	1
0	1	1	1
1	0	0	0
1	1	0	1

3- Cost = 2 + 3 = 5

$f = g$ The two circuits are functionally equivalent

$\overline{x_1} + x_1 \cdot x_2 = \overline{x_1} + x_2$

we can prove that $f = g$ in two ways

1- Truth table

2- Boolean algebra

I - Axioms :

$0 \cdot 0 = 0$

$0 \cdot 1 = 1 \cdot 0 = 0$

$1 \cdot 1 = 1$

$0 + 0 = 0$

$0 + 1 = 1 + 0 = 1$

$1 + 1 = 1$

$\overline{0} = 1$

$\overline{1} = 0$

II - One variable

$x \cdot 0 = 0$

$x + 0 = x$

$x \cdot 1 = x$

$x + 1 = 1$

$x \cdot x = x$

$x + x = x$

$x \cdot \overline{x} = 0$

$x + \overline{x} = 1$

$\overline{\overline{x}} = x$

III - Two or three variables

1- $x.y = y.x \xrightarrow{dual} x+y = y+x$

$x.0=0$

dual

$x+1=1$ ✓

2- $x.(y.z) = (x.y).z$

$\hookrightarrow x+(y+z) = (x+y)+z$

3- $x.(y+z) = x.y + x.z$

4- $x + x.y = x \xrightarrow{dual} x.(x+y) = x$

5- $x.y + x.\bar{y} = x \xrightarrow{dual} (x+y).(x+\bar{y}) = x$

6- $\overline{x.y} = \bar{x} + \bar{y}$ De Morgan's Law

$\hookrightarrow \overline{x+y} = \bar{x}.\bar{y}$

7- $x + \bar{x}.y = x + y$

8- $x.y + y.z + \bar{x}.z = x.y + \bar{x}.z$

$\hookrightarrow (x+y).(y+z).(\bar{x}+z) = (x+y).(\bar{x}+z)$

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* principle of duality

Given logical expression:

replace each · with + and vice versa

and " " 0 with 1 " "

you will get a new logical expression

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minimize the following expression using Boolean algebra

$$\bar{x}_1 x_2 + x_1\bar{x}_2 + \bar{x}_1\bar{x}_2 =$$

$$= \bar{x}_1(x_2+\bar{x}_2) + x_1\bar{x}_2$$

$$= \bar{x}_1(1) + x_1\bar{x}_2$$

$$= \bar{x}_1 + x_1\bar{x}_2$$

$$= \bar{x}_1 + \bar{x}_2$$

حل آخر

$$xy + x\bar{y} = x$$

$$\bar{x}_1 x_2 + x_1\bar{x}_2 + \bar{x}_1\bar{x}_2 = \bar{x}_2 + \bar{x}_1 x_2$$

$$= \bar{x}_2 + \bar{x}_1$$

prove that:

$$\underbrace{X + y \cdot z}_{LHS} = \underbrace{(x+y) \cdot (X + z)}_{RHS}$$

$$RHS = (X + y) \cdot (X + z)$$

$$= X \cdot X + X \cdot z + y \cdot X + y \cdot z$$

$$= X + Xz + yX + yz$$

$$= X + yX + yz$$

$$= x + y \cdot z = LHS$$

Example: prove that

$$\underbrace{ab + \bar{a}c + bc}_{LHS} = \underbrace{ab + \bar{a}c}_{RHS}$$

$$LHS = ab + \bar{a}c + bc$$

$$= ab + \bar{a}c + 1.bc$$

$$= ab + \bar{a}c + (a + \bar{a})bc$$

$$= ab + \bar{a}c + abc + \bar{a}bc$$

$$= ab + \bar{a}c = RHS$$

حل آخر

$$\underline{ab+\bar{a}c+bc} = \underline{ab+\bar{a}c}$$

$$LHS = ab+\bar{a}c+bc$$

$$= ab1+\bar{a}1c+1bc$$

$$= ab(c+\bar{c})+\bar{a}(b+\bar{b})c+(a+\bar{a})bc$$

$$= \underline{abc}+ab\bar{c}+\underline{\bar{a}bc}+\bar{a}\bar{b}c+\underline{abc}+\underline{\bar{a}bc}$$

$$= \underline{abc}+\underline{ab\bar{c}}+\underline{\bar{a}bc}+\underline{\bar{a}\bar{b}c}$$

$$RHS = ab+\bar{a}c$$

$$= ab(c+\bar{c})+\bar{a}(b+\bar{b})c$$

$$= \underline{abc}+\underline{ab\bar{c}}+\underline{\bar{a}bc}+\underline{\bar{a}\bar{b}c}$$

$$= LHS$$

prove that:

$$(x_1+x_2+x_3)\cdot(x_1+x_2+\bar{x}_3)\cdot(x_1+\bar{x}_2+x_3)\cdot(x_1+\bar{x}_2+\bar{x}_3)$$
$$\cdot(\bar{x}_1+x_2+x_3)\cdot(\bar{x}_1+x_2+\bar{x}_3)\cdot(x_1+x_2+\bar{x}_3) = x_1 x_2$$

$$LHS = (x_1+x_2)\cdot(\bar{x}_1+x_2)\cdot(x_1+\bar{x}_3)\cdot(x_1+\bar{x}_2+x_3)$$

$$= x_2\cdot(x_1+\bar{x}_3)\cdot(x_1+\bar{x}_2+x_3)$$

$$= x_2\cdot x_1+(\bar{x}_3\cdot\bar{x}_2+x_3)$$

$$= x_2\cdot x_1+(\bar{x}_3\cdot\bar{x}_2)$$

$$= x_2\cdot(x_1+\bar{x}_3)\cdot(x_1+\bar{x}_2)$$

$$\bar{x}_3\cdot\bar{x}_2+x_3$$

$$= x_2\cdot x_1\cdot(x_1+\bar{x}_3)$$

$$= x_2\cdot x_1$$

$$\overline{a\cdot b} = \bar{a}+\bar{b}$$

$$= x_1\cdot x_2 = RHS$$